BEAM DESIGN SUMMARY

Project Name	:	G+2 RB BEAM DESIGN @ LIFT MACHINE ROOM ROOF
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-14.125 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	14.125 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	14.125 m	:	188	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	8.1	6.22	6.47	5.53	2.59	2.51
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	11.02	10.4	9.39
Asv Torsion (sqmm/m)	62.9	59.89	50.95
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No		:	**B2**	
Group No		:	**G2**	
Analysis Reference(Member)	14.125 m	:	186	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	6.24	5.94	9.38	7.64	3.44	1.65
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	119.6
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	16.29	15.67	13.73
Asv Torsion (sqmm/m)	98.96	95.94	86.53
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No		:	**B3**	
Group No		:	**G3**	
Analysis Reference(Member)	14.125 m	:	185	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm

Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	5.46	7.46	7.01	19.27	4.2	15.35
PtClc (%)	0.2	0.2	0.2	0.276	0.2	0.217
AstCalc (sqmm)	119.6	119.6	119.6	164.94	119.6	129.58
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	18.5	12.38	15.29
Asv Torsion (sqmm/m)	119.56	89.84	91.87
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -

Beam No		**:**	**B4**	
Group No		**:**	**G4**	
Analysis Reference(Member)	14.125 m	:	187	
Breadth		:	230	mm
Depth		:	300	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	5.54	7.37	4.87	14.09	2.03	14.63
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.206
AstCalc (sqmm)	119.6	119.6	119.6	119.6	119.6	123.17
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2

Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	16.54	8.96	15.07
Asv Torsion (sqmm/m)	110.04	60.86	90.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	515.59	515.59	515.59
Reinforcement	2L-T8 @ 195	2L-T8 @ 195	2L-T8 @ 195

SFR : -